F78PA Probability & Statistics A

Specimen Midterm test Solutions

1. A factory produces light bulbs from two machines: Machine A and Machine B. Machine A produces 60% of the total output, and Machine B produces the remaining 40%. The probability that a bulb from Machine A is defective is 0.02, while the probability that a bulb from Machine B is defective is 0.05.

   Let the random variable X represent whether a randomly selected bulb is defective ($X = 1$) or not defective ($X = 0$).

   (a) The outcome space, Ω is given by

   $$\Omega = \{\mathrm{A}, \mathrm{B}\} \times \{\mathrm{D}, \mathrm{ND}\} = \{(\mathrm{A}, \mathrm{D}), (\mathrm{A}, \mathrm{ND}), (\mathrm{B}, \mathrm{D}), (\mathrm{B}, \mathrm{ND})\}.$$

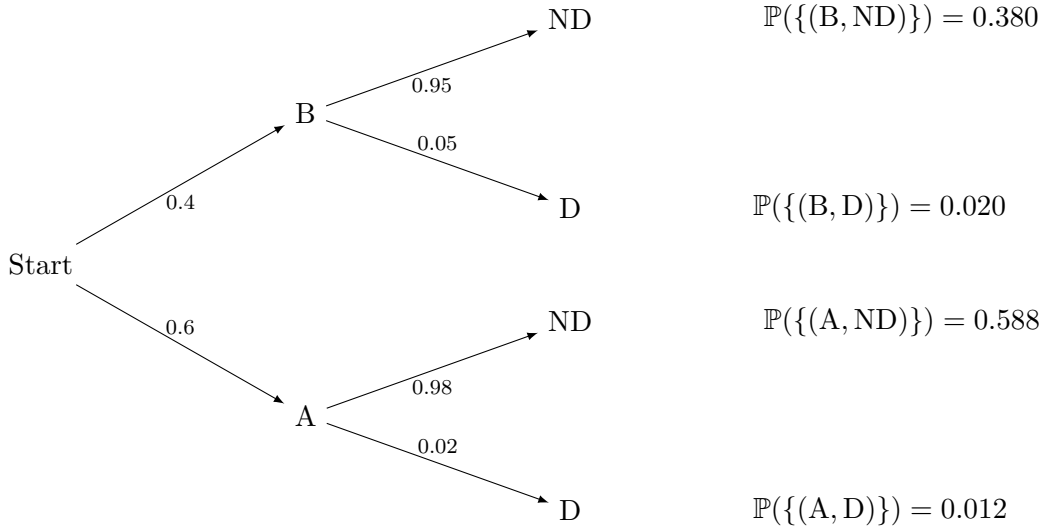
   where A denotes a bulb produced by Machine A, B denotes a bulb produced by Machine B, D denotes a defective bulb and ND is that the bulb is not defective. (1 marks)

   (b) The set $\mathcal{F}$ is a valid event space if

   - The outcome space $\Omega \in \mathcal{F}$;
   - For any $A \in \mathcal{F}$ the complement $A^c = \Omega \setminus A \in \mathcal{F}$;
   - For any countable collection of events $A_1, A_2, \ldots$ the union $\bigcup A_i \in \mathcal{F}$.

   (2 marks)

   (c) The event that the bulb is defective is $\{(A, D), (B, D)\}$, one method to calculate the probability of each outcome is a probability tree such as the one below:



   (1 mark)

   Then

   $$\mathbb{P}(X = 1) = \mathbb{P}(\{(A, D), (B, D)\}) = \mathbb{P}(\{(A, D)\}) + \mathbb{P}(\{(B, D)\}) = 0.012 + 0.02 = 0.032.$$

   (1 mark)

   (d) By the definition of conditional probability

   $$\begin{aligned} \mathbb{P}(\text{Produced by Machine B} | \text{Defective}) &= \frac{\mathbb{P}(\text{Produced by Machine B} \cap \text{Defective})}{\mathbb{P}(\text{Defective})} \\ &= \frac{0.020}{0.032} \\ &= 0.625 \end{aligned}$$

   (2 marks)

(e) The events that the bulb is defective and the bulb is produced by Machine A are independent if

$$\mathbb{P}(\text{Produced by Machine A})\mathbb{P}(\text{Defective}) = \mathbb{P}(\text{Produced by Machine A} \cap \text{Defective})$$

(1 mark for definition of independence)

Using the calculations above

$$\begin{aligned}\mathbb{P}(\text{Produced by Machine A}) &= 0.6\\ \mathbb{P}(\text{Defective}) &= 0.032\\ \mathbb{P}(\text{Produced by Machine A } \cap \text{ Defective}) &= \mathbb{P}(\{(\text{A}, \text{D})\}) = 0.012.\end{aligned}$$

(1 mark)

Therefore since $\mathbb{P}(\text{Produced by Machine A}) \times \mathbb{P}(\text{Defective}) = 0.0192$ which is not equal to $\mathbb{P}(\{(\text{A}, \text{D})\}) = 0.012$ we see that the events that the bulb is defective and the bulb is produced by Machine A are not independent. (1 mark)

[Total 10 marks]

2. A company manufactures metal rods used in construction. The length of a randomly selected rod, measured in metres, is modelled by a continuous random variable X. Quality control analysis suggests that the distribution of rod lengths follows the probability density function:

$$f_X(x) = \begin{cases} kx(2-x), & \text{for } 0 \leq x \leq 2 \\ 0, & \text{otherwise .} \end{cases}$$

(a) A function $f_X : \mathbb{R} \to \mathbb{R}$ is a probability density function for a random variable X if for all $a < b$ we have

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)dx.$$

(1 mark)

(b) The function $f_X(x)$ is a valid probability density function if $f_X(x) \geq 0$ for all x and

$$\int_{-\infty}^{\infty} f_X(x)dx = 1.$$

Since $f_X(x) \geq 0$ for all $k \geq 0$ it remains to determine for what values of k does f_X integrate to 1. Note

$$\begin{aligned}\int_{-\infty}^{\infty} f_X(x)dx &= k\int_0^2 x(2-x)dx\\ &= k\int_0^2 2x - x^2 dx\\ &= k\left[x^2 - \frac{1}{3}x^3\right]_0^2\\ &= k\left(2^2 - \frac{2^3}{3}\right)\\ &= k\left(4 - \frac{8}{3}\right) = \frac{4}{3}k.\end{aligned}$$

Therefore $k = \frac{3}{4}$. (2 marks)

(c) Since $f_X(x) = 0$ for $x < 0$ we have $F_X(x) = 0$ for all $x \leq 0$. (1 mark)
For $x \in (0, 2)$

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{x} f_X(y)dy \\ &= \frac{3}{4}\int_0^x y(2-y)dy \\ &= \frac{3}{4}\left[y^2 - \frac{1}{3}y^3\right]_0^x \\ &= \frac{3}{4}\left(x^2 - \frac{1}{3}x^3\right). \end{aligned}$$

(1 mark)

Then for $x > 2$ we have $F_X(x) = 1$, therefore

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{1}{4}x^2(3-x), & 0 \leq x \leq 2, \\ 1, & x > 2. \end{cases}$$

(1 mark)

(d) The probability that a randomly selected rod has length between 0.5 and 1.5 meters is

$$\mathbb{P}(0.5 \leq X \leq 1.5) = F_X(1.5) - F_X(0.5) = \frac{1}{4}\left(\frac{3}{2}\right)^2\left(3 - \frac{3}{2}\right) - \frac{1}{4}\left(\frac{1}{2}\right)^2\left(3 - \frac{1}{2}\right) = \frac{11}{16}.$$

(1 mark)

(e) Note that $f_X(x)$ is zero outside $[0, 2]$ and is a quadratic function within $[0, 2]$ so it suffices to find the maximum value of $\frac{3}{4}x(2-x)$ for $x \in [0, 2]$. We can complete the square to write

$$\frac{3}{4}x(2-x) = -\frac{3}{4}\left(x^2 - 2x\right) = -\frac{3}{4}\left((x-1)^2 - 1\right).$$

This function is maximised when $x = 1$, so the mode of X is 1. (3 marks)

[Total 10 marks]